

BRAEDEN KEENA

Data Engineer · Recurring Pipelines · Data Integrity & Failure Detection

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braedenkeena.pages.dev · github.com/itshumpf · gascout.pages.dev · nacup.us

I build data collection that keeps working, and I specialize in the failure that looks like success — a pipeline reporting a clean run while returning nothing. Every figure I publish is emitted by a script that refuses to write a null. Backed by front-line operations leadership, including founding and running my own business, managing teams of up to 100 people.

SHIPPED PRODUCTS (SOLO-BUILT, END TO END)

WiFi CSI RF-Fingerprinting Sensor Array — ESP32 + Python DSP *54 days, from zero · Firmware / DSP*

- Fifty-four days from not knowing what an ESP32 was to a closed scientific loop — instrument, method, answer, next hypothesis. Built on a repurposed Meshtastic node and a beginner kit off Amazon: five boards, C / ESP-IDF firmware, and a Python DSP chain (RANSAC phase-slope fits, Kalman drift tracking, Mahalanobis discrimination).
- Restarted the project in July specifically to test it rather than defend it. Two months and twenty-nine hypotheses carried to a verdict, thirteen of them my own working results, each retired by a control I built to permit that outcome.
- The separation was real. What it was separating wasn't. Two boards four feet apart read 0.2σ ; one moved between rooms read 19.6σ — rooms, not radios. Then 69.5 million frames across eight paired overnight captures, scored against decision rules frozen before the run, showed it was neither: the transmitter, the receiver, and the path between them, measured as one object.
- It still identifies three co-located transmitters at 95.8% blind accuracy — it just isn't a property of the radio alone. Every threshold written to disk and dated before the test that used it, results that went against me published beside the ones that didn't. V2: find the link, measure it, find out whether it can be tracked.
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findstorage.pages.dev — Self-Storage Price Tracker *2025 – 2026 · Founder / Developer · Sunset by choice*

- Daily national advertised-price tracker: 4,664 stores across 41 states plus DC and Puerto Rico, 46 consecutive daily snapshots with zero gaps, 396,402 price changes and 148,870 promotion changes logged. 7-pass discovery scraper with safety rails that abort a publish if the store count drops below a floor or swings $>10\%$ run-over-run — an operator merger added $\sim 1,100$ stores overnight and the pipeline absorbed it with no manual intervention.
- Shut the project down on 25 August 2026 after re-reading the source's terms of use, and published why rather than stopping quietly — killing a six-week analysis that a lawyer had already cleared, because the data underneath it was collected under terms I had not re-checked.
- Found and fixed four separate errors in my own published analysis before shutting down, including a coverage figure wrong by $5.6\times$ that no script had ever recomputed. Every published number now traces to a named command.

Public-Records Extraction — GAScout · easyIce · KcPermits

2025 – 2026 · Solo-built

- Turn government systems never designed to be read programmatically into structured, deduplicated, ranked datasets. GAScout (gascout.pages.dev): \$152.34M in delinquent taxes and 451,798 records across four live Georgia counties in four source formats — a 9,915-page mainframe PDF, a weekly XLSX roll, a monthly PDF, and a scanned PDF read by OCR — normalized to one schema.
- easyIce converts county health-inspection records into a recency-scored call list across five jurisdiction adapters and three source shapes (Socrata JSON API, server-rendered HTML with sequential IDs, Tyler portals); 387 tests, none touching the network. Every run publishes a coverage report that declares its own degradation first — a partial run cannot be mistaken for a complete one.
- KcPermits consolidates 57,032 records from 21 municipal and county feeds into 30,587 resolved locations. Reverse-engineered Tyler EnerGov's four undocumented tenant headers across four city portals — tenant IDs differ between identical vendor installs — then measured two silent failures: a 10,000-result ceiling that returns HTTP 200 with no warning, and a date filter ignored server-side, where a 90-day window returned the same 124,214 rows as no window at all.
- The deliverable is built to refuse its own output: 37 source records consolidate to 10 projects and 4 prospects, each published with the limitation requiring resolution before action. A written spec constrains the renderer to fields that exist in a record, so the presentation layer cannot compose a claim the data does not carry.

nacup.us — Live World Cup 2026 Fan Hub

2026 · Founder / Developer

- Designed, built, and deployed a full-stack production site in 48 hours; reached first-page organic search traffic within 24 hours of launch with 700+ visitors on day one during a live global event.
- Multi-source data pipeline (Google Places API, RSS, news scraping, live scores REST API) with dedupe/merge logic, trust-tier scoring, and freshness decay ranking across 38 host-city feeds.
- Programmatically generated 185 static SEO pages with JSON-LD structured data, sitemap, and Search Console integration; live-score polling with caching and fallback states.

TECHNICAL SKILLS

- **Languages:** Python (scraping, data pipelines, automation), SQL, JavaScript, C, HTML/CSS
- **Data:** ETL pipelines, entity resolution, deduplication, pandas, SQLite, DuckDB, Parquet, REST APIs, API reverse-engineering, Playwright, PDF and OCR extraction (Tesseract, pdfplumber), JSON data architecture
- **Testing & Reliability:** pytest, fixture-based contract tests, negative controls, coverage reporting, pre-registered thresholds, failure detection and alerting
- **DevOps:** Git/GitHub, CI/CD (GitHub Actions, Cloudflare Pages, Netlify), scheduled jobs, robots.txt and rate-limit compliance
- **Hardware/IoT:** ESP32 firmware (C / ESP-IDF), ESP-NOW & BLE, I2C peripherals, sensor networks, 3D printing & CAD (OpenSCAD)

OPERATIONS & LEADERSHIP EXPERIENCE

Front-line operations and team leadership — including founding and operating my own business — managing teams of up to 100 people across high-volume service environments, staffing, and vendor coordination.

Independent Technical Consultant

7/2026 – Present

- Build and ship production scrapers, data pipelines, and automation — turning messy public and web data sources into clean, scheduled, queryable datasets and dashboards (see shipped projects above).
- Contract work delivered across records consolidation and data visualization; 100% Job Success Score and Rising Talent status on Upwork as of September 2026.

Crossing Guard Operations Supervisor — All City Management Services (ACMS) 2023 – 2025

- Hired, trained, and managed 50 crossing guards across 33 sites — 66 posts staffed twice every school day — holding ~25% annual turnover in a category known for severe churn.
- Coordinated daily across five constituencies (county sheriff, school principals, parents, the district, and the guards) and solved same-day coverage through a deep bench of on-call relationships — every student returned home safely, zero students injured across my full tenure.

Founder / Sole Operating Partner — Honey Lake Motocross Park, Milford, CA 2016 – 2018

- Acquired and revived a ~500-acre desert motocross park dormant since 2007 — bought with two capital partners as the sole operating owner — and re-entitled it for racing after the prior owner had lost county approval and stripped the site: new survey, environmental review, a water well drilled under budget, and a heavy-equipment track rebuild.
- Directed a 120-person race-day crew built largely on in-kind pay with no payroll budget (peak ~6,500 attendees); designed and staffed the on-site medical program that executed a mid-event helicopter evacuation without incident.
- Held full P&L — pricing, sponsorship, staffing, and marketing — and founded the Corey Herring Memorial race, which continued annually through 2023; exited by selling my stake.

EDUCATION

GED — State of Kansas, 2014. Everything since: self-taught, project-driven, and shipped to production.